

District 5 SPSS Comparison 2023 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

13 May 2026

Contents

Model I	1
Model II	2
Model III	4
Model IV	5
Model V	7
Model VI	9

Model I

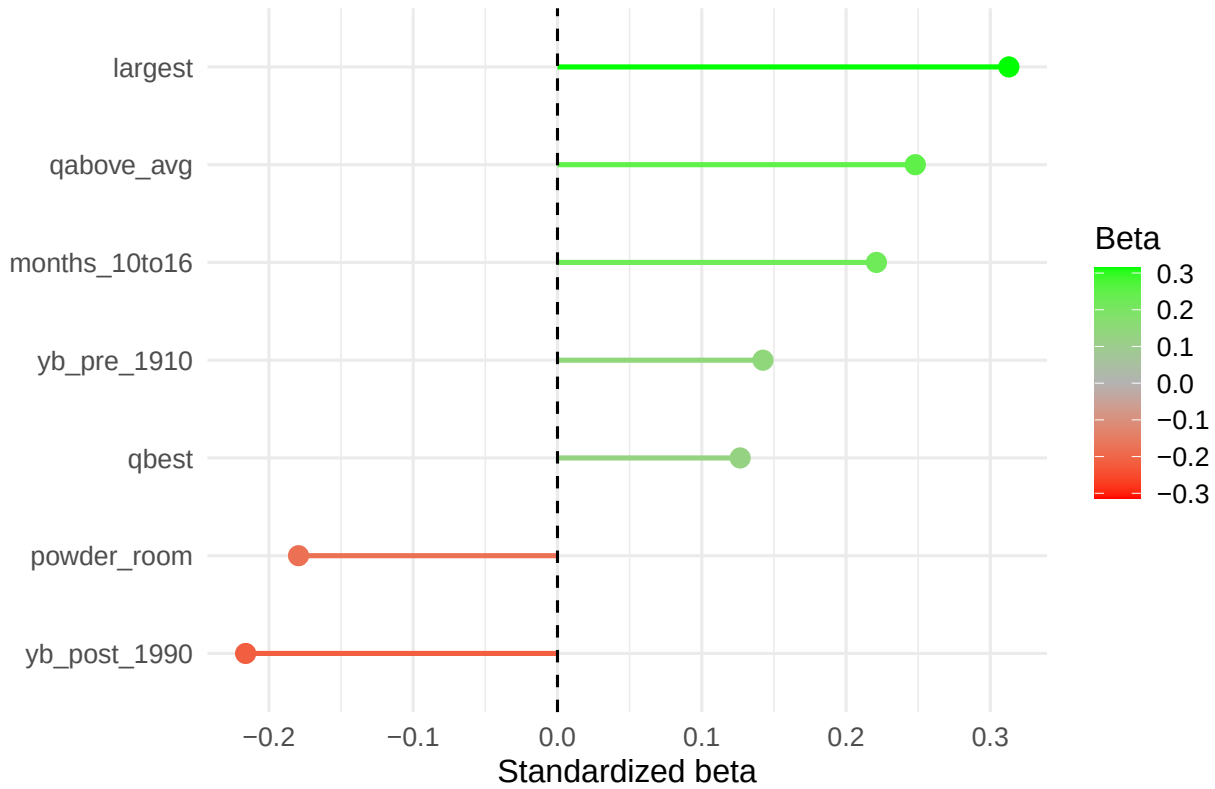
Table 1: District 5 | LN_SPPSF Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	160	153	7	0.6275	0.3930	0.3650	14.1390
R	160	153	7	0.6275	0.3928	0.3650	14.1386
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.01%	-0.05%	0.00%	-0.00%

Table 2: District 5 | Model I Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	3.825	NA	0.104	NA	NA	NA	36.634	NA	0.000	NA
largest	0.492	NA	0.132	NA	0.313	NA	3.726	NA	0.000	NA
qabove_avg	0.447	NA	0.140	NA	0.248	NA	3.197	NA	0.002	NA
months_10to16	0.055	NA	0.016	NA	0.221	NA	3.479	NA	0.001	NA
yb_pre_1910	0.410	NA	0.186	NA	0.142	NA	2.207	NA	0.029	NA
qbest	0.897	NA	0.456	NA	0.127	NA	1.968	NA	0.051	NA
powder_room	-0.282	NA	0.109	NA	-0.179	NA	-2.585	NA	0.011	NA
yb_post_1990	-0.896	NA	0.270	NA	-0.216	NA	-3.320	NA	0.001	NA

District 5 | LN_SPPSF Model Standardized Beta Values



Model II

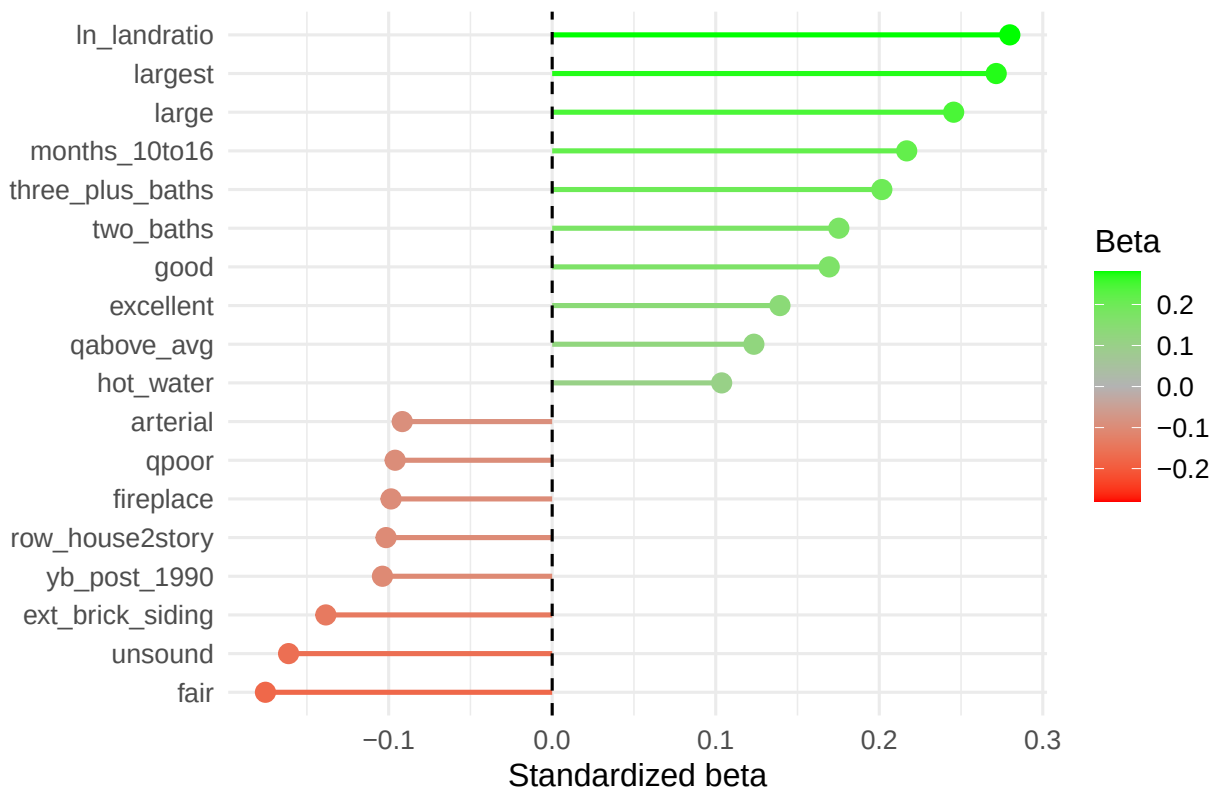
Table 3: District 5 | LN_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	307	289	18	0.6744	0.5350	0.5060	18.4470
R	307	289	18	0.6744	0.5346	0.5057	18.4465
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.01%	-0.07%	-0.06%	-0.00%

Table 4: District 5 | Model II Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	10.822	10.822	0.139	0.139	NA	NA	77.753	77.753	0.000	0.001
ln_landratio	0.677	0.677	0.144	0.144	0.280	0.280	4.693	4.693	0.000	0.001
largest	0.523	0.523	0.170	0.170	0.271	0.271	3.077	3.077	0.002	0.002
large	0.493	0.493	0.147	0.147	0.245	0.245	3.350	3.350	0.001	0.001
months_10to16	0.067	0.067	0.013	0.013	0.217	0.217	5.144	5.144	0.000	0.001
three_plus_baths	0.661	0.661	0.197	0.197	0.202	0.202	3.356	3.356	0.001	0.001
two_baths	0.364	0.364	0.130	0.130	0.175	0.175	2.804	2.804	0.005	0.005
good	0.583	0.583	0.145	0.145	0.169	0.169	4.014	4.014	0.000	0.001
excellent	1.358	1.358	0.432	0.432	0.139	0.139	3.144	3.144	0.002	0.002
qabove_avg	0.275	0.275	0.127	0.127	0.123	0.123	2.169	2.169	0.031	0.031
hot_water	0.199	0.199	0.097	0.097	0.104	0.104	2.053	2.053	0.041	0.041
arterial	-0.273	-0.273	0.125	0.125	-0.092	-0.092	-2.187	-2.187	0.030	0.030
qpoor	-1.617	-1.617	0.724	0.724	-0.096	-0.096	-2.234	-2.234	0.026	0.026
fireplace	-0.197	-0.197	0.097	0.097	-0.099	-0.099	-2.033	-2.033	0.043	0.043
row_house2story	-1.710	-1.710	0.712	0.712	-0.102	-0.102	-2.403	-2.403	0.017	0.017
yb_post_1990	-0.590	-0.590	0.252	0.252	-0.104	-0.104	-2.346	-2.346	0.020	0.020
ext_brick_siding	-0.337	-0.337	0.111	0.111	-0.138	-0.138	-3.046	-3.046	0.003	0.003
unsound	-1.922	-1.922	0.491	0.491	-0.161	-0.161	-3.915	-3.915	0.000	0.001
fair	-0.374	-0.374	0.090	0.090	-0.175	-0.175	-4.144	-4.144	0.000	0.001

District 5 | LN_PRICE (– ECF) Model Std Beta Values



Model III

Table 5: District 5 | LN_PRICE Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	307	281	26	0.5948	0.6480	0.6150	19.8960
R	307	281	26	0.5948	0.6480	0.6154	19.8957
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.00%	0.00%	0.07%	-0.00%

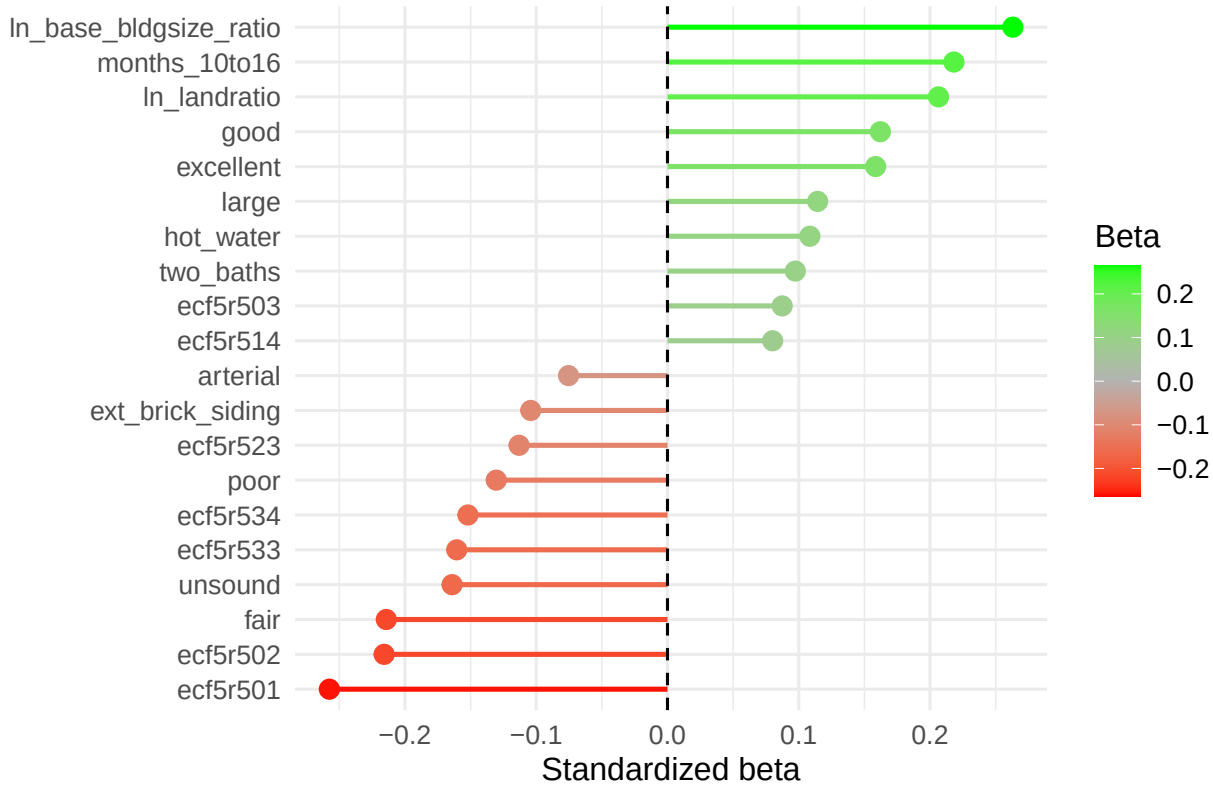
Table 6: District 5 | LN_PRICE Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0000	1.2930	0.9090	1.4230	0.6450	
R	0.9803	1.1826	0.9003	1.3136	0.5156	-0.2192
% diff (R vs SPSS)	-1.97%	-8.54%	-0.96%	-7.69%	-20.07%	

Table 7: District 5 | Model III Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.428	11.428	0.114	0.114	NA	NA	100.207	100.207	0.000	0.001
ln_base_bldgsize_ratio	0.747	0.747	0.155	0.155	0.263	0.263	4.826	4.826	0.000	0.001
months_10to16	0.067	0.067	0.012	0.012	0.218	0.218	5.761	5.761	0.000	0.001
ln_landratio	0.500	0.500	0.126	0.126	0.206	0.206	3.959	3.959	0.000	0.001
good	0.559	0.559	0.130	0.130	0.162	0.162	4.299	4.299	0.000	0.001
excellent	1.546	1.546	0.393	0.393	0.159	0.159	3.930	3.930	0.000	0.001
large	0.230	0.230	0.099	0.099	0.114	0.114	2.314	2.314	0.021	0.021
hot_water	0.208	0.208	0.085	0.085	0.108	0.108	2.466	2.466	0.014	0.014
two_baths	0.203	0.203	0.090	0.090	0.097	0.097	2.248	2.248	0.025	0.025
ecf5r503	0.472	0.472	0.214	0.214	0.087	0.087	2.210	2.210	0.028	0.028
ecf5r514	0.346	0.346	0.167	0.167	0.080	0.080	2.072	2.072	0.039	0.039
ecf5r517_521	0.504	0.504	0.243	0.243	0.078	0.078	2.077	2.077	0.039	0.039
ecf5r529	0.329	0.329	0.163	0.163	0.078	0.078	2.022	2.022	0.044	0.044
ecf5r530_532_535	0.368	0.368	0.204	0.204	0.071	0.071	1.806	1.806	0.072	0.072
ecf5r538	-1.047	-1.047	0.612	0.612	-0.062	-0.062	-1.712	-1.712	0.088	0.088
row_house2story	-1.109	-1.109	0.626	0.626	-0.066	-0.066	-1.770	-1.770	0.078	0.078
ecf5r528	-0.576	-0.576	0.307	0.307	-0.068	-0.068	-1.875	-1.875	0.062	0.062
arterial	-0.225	-0.225	0.116	0.116	-0.075	-0.075	-1.930	-1.930	0.055	0.055
ext_brick_siding	-0.253	-0.253	0.100	0.100	-0.104	-0.104	-2.534	-2.534	0.012	0.012
ecf5r523	-0.727	-0.727	0.242	0.242	-0.113	-0.113	-3.006	-3.006	0.003	0.003
poor	-0.421	-0.421	0.128	0.128	-0.130	-0.130	-3.297	-3.297	0.001	0.001
ecf5r534	-0.621	-0.621	0.171	0.171	-0.152	-0.152	-3.635	-3.635	0.000	0.001
ecf5r533	-0.913	-0.913	0.231	0.231	-0.161	-0.161	-3.955	-3.955	0.000	0.001
unsound	-1.956	-1.956	0.436	0.436	-0.164	-0.164	-4.483	-4.483	0.000	0.001
fair	-0.457	-0.457	0.082	0.082	-0.214	-0.214	-5.598	-5.598	0.000	0.001
ecf5r502	-0.536	-0.536	0.109	0.109	-0.216	-0.216	-4.924	-4.924	0.000	0.001
ecf5r501	-0.631	-0.631	0.107	0.107	-0.258	-0.258	-5.909	-5.909	0.000	0.001

District 5 | LN_PRICE Model Standardized Beta Values



Model IV

Table 8: District 5 | LN_PRICE (Inliers) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	262	238	24	0.4466	0.7790	0.7570	35.0370
R	260	237	23	0.4452	0.7767	0.7550	35.8318
% diff (R vs SPSS)	-0.76%	-0.42%	-4.17%	-0.32%	-0.30%	-0.26%	2.27%

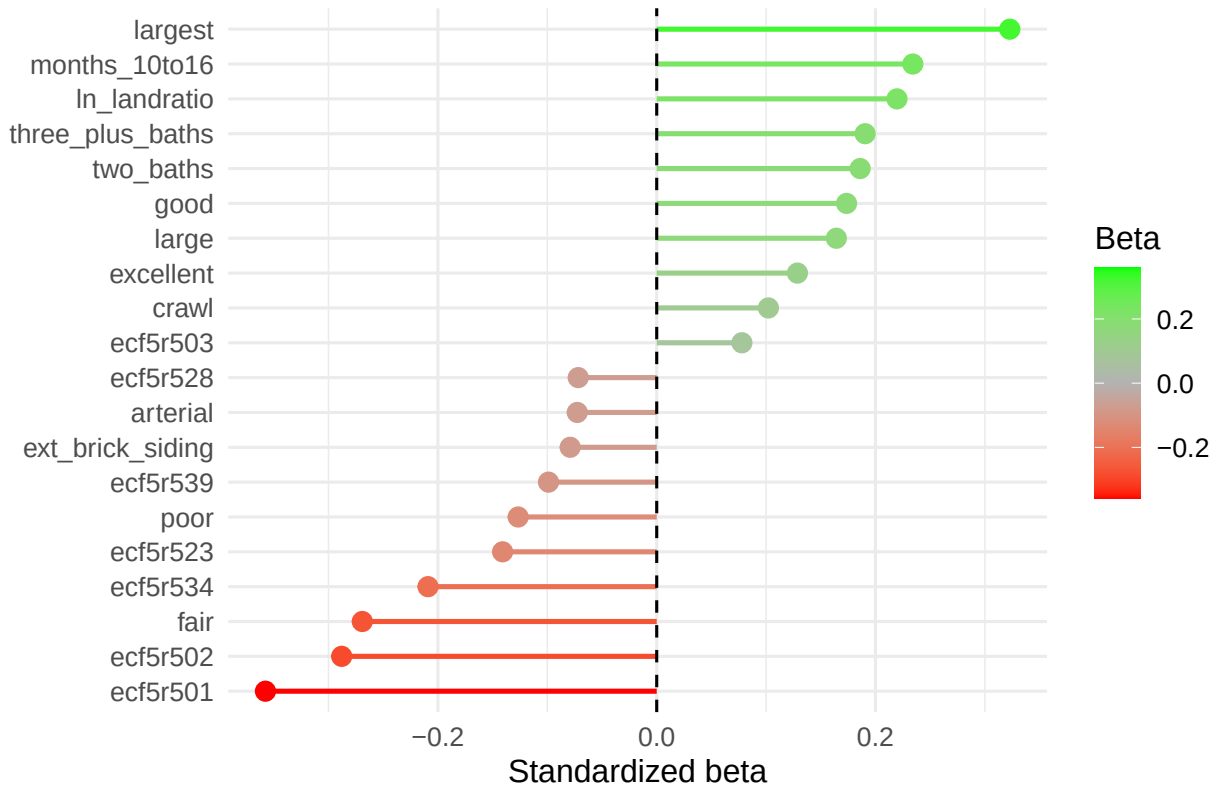
Table 9: District 5 | LN_PRICE (Inliers) Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1	1.1470	0.9440	1.2150	0.4370	-0.1290
R	1	1.0948	0.9384	1.1667	0.3630	-0.1106
% diff (R vs SPSS)	-0.00%	-4.55%	-0.60%	-3.98%	-16.93%	14.28%

Table 10: District 5 | Model IV Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.337	11.319	0.112	0.112	NA	NA	101.308	101.349	0.000	0.001
largest	0.588	0.580	0.121	0.121	0.323	0.316	4.874	4.791	0.000	0.001
months_10to16	0.068	0.066	0.010	0.009	0.234	0.225	7.103	6.924	0.000	0.001
ln_landratio	0.497	0.478	0.105	0.104	0.220	0.210	4.744	4.575	0.000	0.001
three_plus_baths	0.562	0.513	0.141	0.138	0.190	0.172	3.991	3.723	0.000	0.001
two_baths	0.359	0.315	0.092	0.088	0.186	0.163	3.885	3.564	0.000	0.001
good	0.586	0.586	0.110	0.111	0.174	0.175	5.305	5.287	0.000	0.001
large	0.314	0.306	0.111	0.112	0.164	0.159	2.813	2.737	0.005	0.007
excellent	1.869	1.720	0.488	0.480	0.129	0.117	3.832	3.583	0.000	0.001
crawl	0.746	0.742	0.297	0.298	0.102	0.100	2.514	2.493	0.013	0.013
ecf5r503	0.406	0.354	0.180	0.177	0.078	0.067	2.256	1.994	0.025	0.047
ecf5r517_521	0.380	0.401	0.238	0.238	0.052	0.054	1.598	1.683	0.111	0.094
qabove_avg	-0.141	NA	0.089	NA	-0.068	NA	-1.589	NA	0.113	NA
ecf5r533	-0.714	-0.699	0.343	0.344	-0.069	-0.067	-2.084	-2.035	0.038	0.043
ecf5r528	-0.738	-0.692	0.322	0.322	-0.072	-0.066	-2.292	-2.151	0.023	0.032
arterial	-0.204	-0.205	0.098	0.098	-0.073	-0.072	-2.091	-2.088	0.038	0.038
ext_brick_siding	-0.180	-0.163	0.078	0.078	-0.079	-0.071	-2.310	-2.103	0.022	0.037
ecf5r539	-0.833	-0.797	0.284	0.284	-0.099	-0.094	-2.936	-2.809	0.004	0.005
poor	-0.401	-0.387	0.106	0.106	-0.127	-0.121	-3.781	-3.652	0.000	0.001
ecf5r523	-1.186	-1.163	0.330	0.331	-0.141	-0.137	-3.590	-3.512	0.000	0.001
ecf5r534	-0.978	-0.934	0.162	0.160	-0.209	-0.197	-6.030	-5.827	0.000	0.001
fair	-0.534	-0.519	0.066	0.065	-0.269	-0.260	-8.146	-7.974	0.000	0.001
ecf5r502	-0.657	-0.628	0.084	0.082	-0.288	-0.272	-7.819	-7.632	0.000	0.001
ecf5r501	-0.782	-0.754	0.081	0.080	-0.358	-0.341	-9.598	-9.450	0.000	0.001
ecf5r538	NA	-1.121	NA	0.464	NA	-0.076	NA	-2.417	NA	0.016
row_house2story	NA	-1.756	NA	0.466	NA	-0.119	NA	-3.772	NA	0.001

District 5 | LN_PRICE (Inliers) Model Std Beta Values



Model V

Table 11: District 5 | LN_TASP Model I Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	262	239	23	0.4458	0.7740	0.7520	35.5210
R	260	239	21	0.4473	0.7671	0.7466	37.4852
% diff (R vs SPSS)	-0.76%	0.00%	-8.70%	0.35%	-0.89%	-0.72%	5.53%

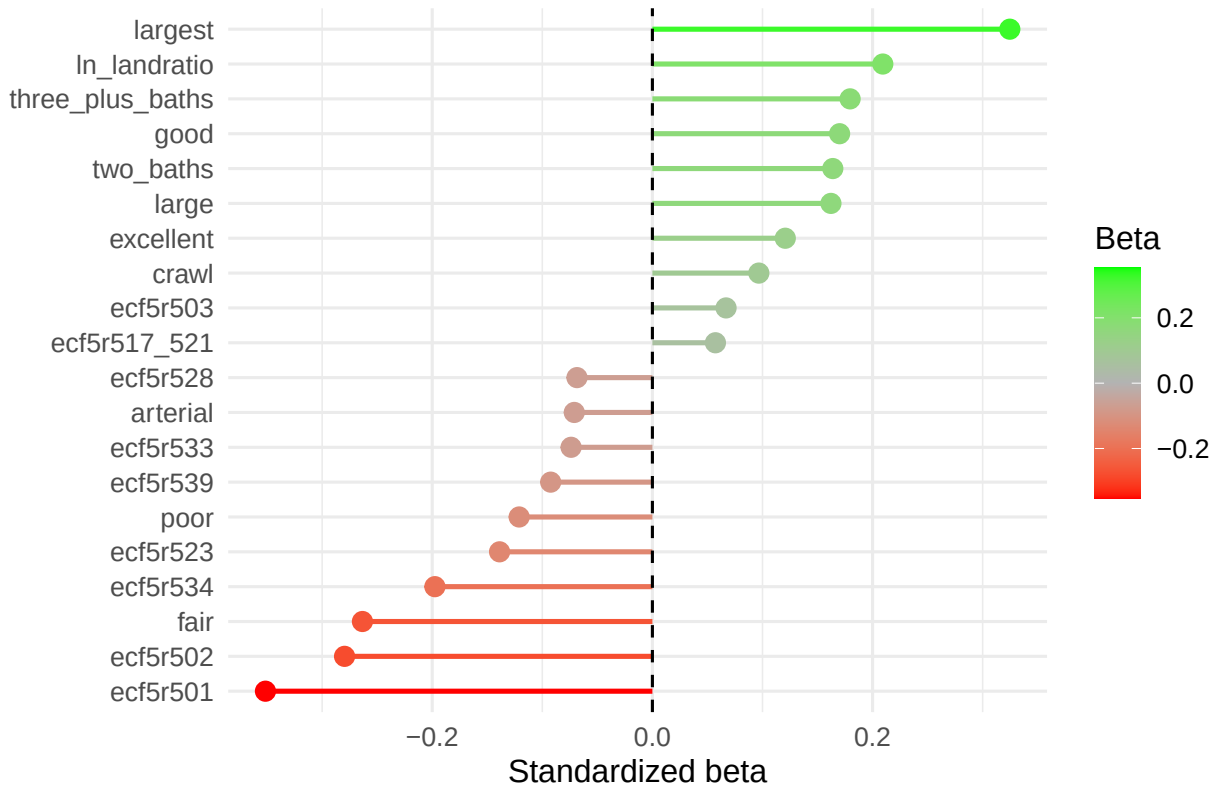
Table 12: District 5 | LN_TASP Model I Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0020	1.1460	0.9430	1.2150	0.4360	-0.1300
R	1.1449	1.3180	1.0973	1.2011	0.4232	-0.1370
% diff (R vs SPSS)	14.27%	15.01%	16.37%	-1.14%	-2.94%	-5.42%

Table 13: District 5 | Model V Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.736	11.771	0.107	0.106	NA	NA	110.175	110.887	0.000	0.001
largest	0.585	0.580	0.121	0.121	0.325	0.320	4.829	4.810	0.000	0.001
ln_landratio	0.468	0.476	0.104	0.104	0.209	0.212	4.488	4.580	0.000	0.001
three_plus_baths	0.523	0.515	0.138	0.137	0.180	0.175	3.797	3.749	0.000	0.001
good	0.567	0.582	0.110	0.110	0.170	0.177	5.155	5.313	0.000	0.001
two_baths	0.312	0.315	0.089	0.088	0.164	0.164	3.524	3.567	0.001	0.001
large	0.306	0.306	0.112	0.112	0.162	0.161	2.735	2.743	0.007	0.007
excellent	1.733	1.722	0.481	0.479	0.121	0.119	3.605	3.595	0.000	0.001
crawl	0.698	0.735	0.296	0.295	0.097	0.101	2.357	2.489	0.019	0.013
ecf5r503	0.344	0.352	0.177	0.177	0.067	0.068	1.940	1.991	0.054	0.048
ecf5r517_521	0.413	0.403	0.238	0.238	0.057	0.055	1.735	1.696	0.084	0.091
ext_brick_siding	-0.147	-0.160	0.077	0.076	-0.065	-0.070	-1.913	-2.099	0.057	0.037
ecf5r528	-0.697	-0.693	0.322	0.321	-0.069	-0.067	-2.162	-2.158	0.032	0.032
arterial	-0.197	-0.204	0.098	0.098	-0.071	-0.072	-2.013	-2.083	0.045	0.038
ecf5r533	-0.751	-0.708	0.342	0.340	-0.074	-0.069	-2.199	-2.078	0.029	0.039
ecf5r539	-0.769	-0.792	0.283	0.282	-0.092	-0.094	-2.714	-2.806	0.007	0.005
poor	-0.379	-0.386	0.106	0.106	-0.121	-0.122	-3.572	-3.652	0.000	0.001
ecf5r523	-1.155	-1.162	0.332	0.330	-0.139	-0.138	-3.483	-3.515	0.001	0.001
ecf5r534	-0.913	-0.930	0.160	0.159	-0.198	-0.199	-5.716	-5.847	0.000	0.001
fair	-0.516	-0.518	0.065	0.065	-0.263	-0.263	-7.929	-7.987	0.000	0.001
ecf5r502	-0.630	-0.629	0.082	0.082	-0.280	-0.276	-7.647	-7.652	0.000	0.001
ecf5r501	-0.759	-0.755	0.080	0.079	-0.351	-0.346	-9.523	-9.496	0.000	0.001
ecf5r538	NA	-1.129	NA	0.461	NA	-0.078	NA	-2.448	NA	0.015
row_house2story	NA	-1.745	NA	0.462	NA	-0.120	NA	-3.778	NA	0.001

District 5 | LN_TASP Model I Standardized Beta Values



Model VI

Table 14: District 5 | LN_TASP Model II Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	505	477	28	0.4578	0.7440	0.7290	49.5400
R	260	239	21	0.4473	0.7671	0.7466	37.4852
% diff (R vs SPSS)	-48.51%	-49.90%	-25.00%	-2.30%	3.10%	2.41%	-24.33%

Table 15: District 5 | LN_TASP Model II Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0000	1.1030	0.9290	1.1880	0.3790	-0.1290
R	1.1449	1.3180	1.0973	1.2011	0.4232	-0.1370
% diff (R vs SPSS)	14.49%	19.49%	18.12%	1.10%	11.65%	-6.24%

Table 16: District 5 | Model VI Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.736	9.796	0.107	0.552	NA	NA	110.175	17.732	0.000	0.001
largest	0.585	NA	0.121	NA	0.325	NA	4.829	NA	0.000	NA
ln_landratio	0.468	0.445	0.104	0.081	0.209	0.186	4.488	5.521	0.000	0.001
three_plus_baths	0.523	0.261	0.138	0.131	0.180	0.073	3.797	1.991	0.000	0.047
good	0.567	0.689	0.110	0.080	0.170	0.214	5.155	8.632	0.000	0.001
two_baths	0.312	0.170	0.089	0.067	0.164	0.083	3.524	2.537	0.001	0.012
large	0.306	NA	0.112	NA	0.162	NA	2.735	NA	0.007	NA
excellent	1.733	1.604	0.481	0.481	0.121	0.081	3.605	3.335	0.000	0.001
crawl	0.698	NA	0.296	NA	0.097	NA	2.357	NA	0.019	NA
ecf5r503	0.344	0.574	0.177	0.175	0.067	0.086	1.940	3.285	0.054	0.001
ecf5r517_521	0.413	0.593	0.238	0.142	0.057	0.103	1.735	4.183	0.084	0.001
ext_brick_siding	-0.147	-0.174	0.077	0.060	-0.065	-0.073	-1.913	-2.887	0.057	0.004
ecf5r528	-0.697	-0.475	0.322	0.127	-0.069	-0.097	-2.162	-3.729	0.032	0.001
arterial	-0.197	-0.173	0.098	0.066	-0.071	-0.065	-2.013	-2.602	0.045	0.010
ecf5r533	-0.751	-0.739	0.342	0.132	-0.074	-0.143	-2.199	-5.585	0.029	0.001
ecf5r539	-0.769	NA	0.283	NA	-0.092	NA	-2.714	NA	0.007	NA
poor	-0.379	-0.436	0.106	0.076	-0.121	-0.141	-3.572	-5.760	0.000	0.001
ecf5r523	-1.155	-0.549	0.332	0.172	-0.139	-0.078	-3.483	-3.196	0.001	0.001
ecf5r534	-0.913	-0.501	0.160	0.123	-0.198	-0.106	-5.716	-4.082	0.000	0.001
fair	-0.516	-0.487	0.065	0.047	-0.263	-0.257	-7.929	-10.304	0.000	0.001
ecf5r502	-0.630	-0.449	0.082	0.072	-0.280	-0.185	-7.647	-6.262	0.000	0.001
ecf5r501	-0.759	-0.604	0.080	0.072	-0.351	-0.263	-9.523	-8.413	0.000	0.001
ecf5r504	NA	0.229	NA	0.097	NA	0.065	NA	2.355	NA	0.019
ecf5r508_509	NA	0.265	NA	0.143	NA	0.046	NA	1.851	NA	0.065
ecf5r514	NA	0.449	NA	0.089	NA	0.133	NA	5.019	NA	0.001
ecf5r529	NA	0.534	NA	0.084	NA	0.181	NA	6.371	NA	0.001
ecf5r530_532_535	NA	0.481	NA	0.147	NA	0.083	NA	3.284	NA	0.001
ecf5r538	NA	-0.876	NA	0.197	NA	-0.108	NA	-4.436	NA	0.001
fireplace	NA	0.113	NA	0.057	NA	0.064	NA	1.982	NA	0.048
hot_water	NA	0.226	NA	0.048	NA	0.128	NA	4.683	NA	0.001
lnsbldsfint	NA	0.287	NA	0.081	NA	0.149	NA	3.532	NA	0.001
row_house2story	NA	-1.163	NA	0.472	NA	-0.059	NA	-2.465	NA	0.014
unsound	NA	-1.437	NA	0.474	NA	-0.073	NA	-3.032	NA	0.003

District 5 | LN_TASP Model II Standardized Beta Values

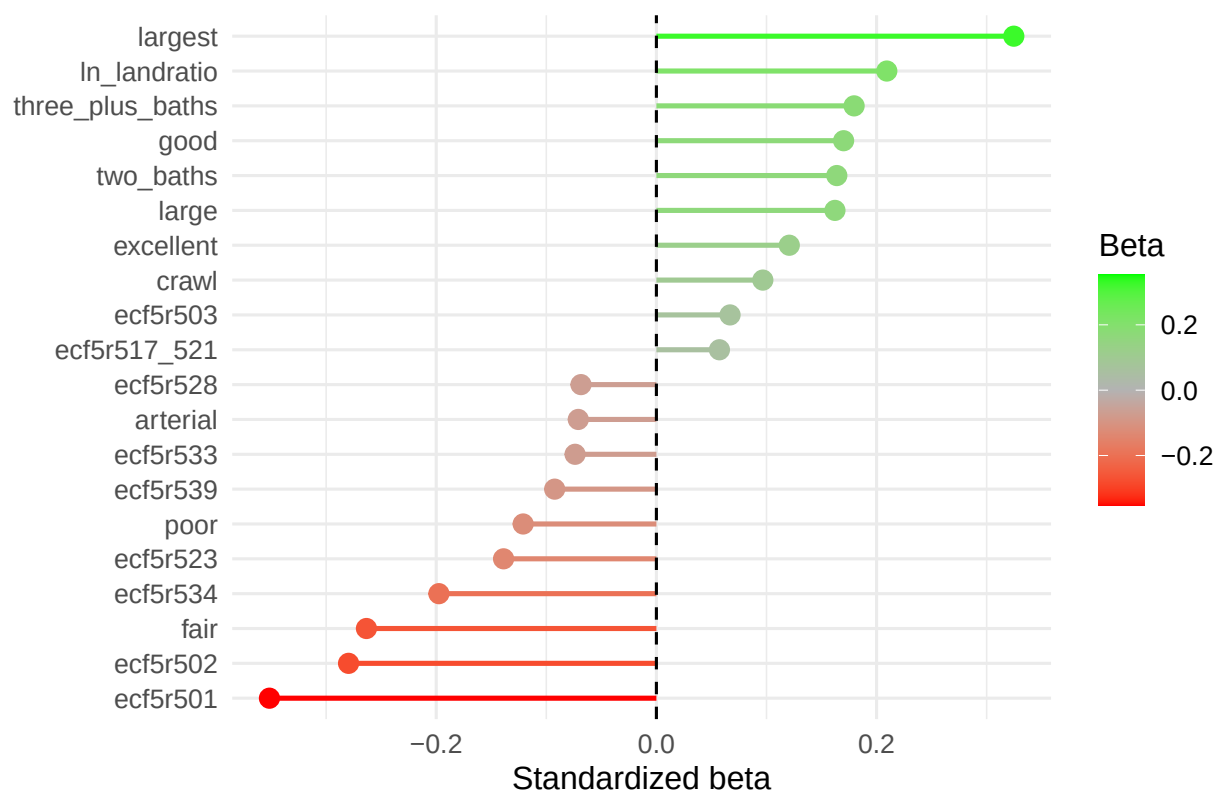


Table 17: Top and Bottom Standardized Beta Predictors across Models

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
largest	5	M1, M2, M4, M5, M6	0.555	0.133	0.311	4.267	0.001	0.271	0.325
ln_base_bldgsize_ratio	1	M3	0.747	0.155	0.263	4.826	0.000	0.263	0.263
qabove_avg	1	M1	0.447	0.140	0.248	3.197	0.002	0.248	0.248
large	1	M2	0.493	0.147	0.245	3.350	0.001	0.245	0.245
ln_landratio	5	M2, M3, M4, M5, M6	0.522	0.117	0.225	4.475	0.000	0.206	0.280
months_10to16	3	M1, M3, M4	0.063	0.012	0.224	5.448	0.000	0.218	0.234
three_plus_baths	2	M5, M6	0.523	0.138	0.180	3.797	0.000	0.180	0.180
qbest	1	M1	0.897	0.456	0.127	1.968	0.051	0.127	0.127
ext_brick_siding	1	M2	-0.337	0.111	-0.138	-3.046	0.003	-0.138	-0.138
unsound	1	M2	-1.922	0.491	-0.161	-3.915	0.000	-0.161	-0.161
powder_room	1	M1	-0.282	0.109	-0.179	-2.585	0.011	-0.179	-0.179
yb_post_1990	1	M1	-0.896	0.270	-0.216	-3.320	0.001	-0.216	-0.216
fair	5	M2, M3, M4, M5, M6	-0.480	0.074	-0.237	-6.749	0.000	-0.269	-0.175
ecf5r502	4	M3, M4, M5, M6	-0.613	0.089	-0.266	-7.009	0.000	-0.288	-0.216
ecf5r501	4	M3, M4, M5, M6	-0.733	0.087	-0.330	-8.638	0.000	-0.358	-0.258

District 5 | Average Standardized Beta across Models

